

Konsider Phase 6B.1 Prompt — Career Opportunity Gap Fill and Threshold Calibration

Role

Work as the lead product, labour-data, evidence, and reproducibility engineer for the Konsider repository.

This is a focused follow-up to Phase 6B.

Do not implement production schemas, workers, releases, ranking changes, API changes, UI changes, presets, conversational behavior, or applicant-profile logic.

Temporary research scripts, retained local source captures, deterministic mappings, fixtures, and committed research reports are allowed.

Starting point

Begin from the latest `main` branch and inspect the working tree before making changes.

Read at minimum:

- `docs/research/phase6b-career-opportunity-study.md`
- `data/reports/phase6b-2026-08-02/approved-career-opportunity-portfolio.json`
- `data/reports/phase6b-2026-08-02/career-candidate-matrix.json`
- `data/reports/phase6b-2026-08-02/career-country-opportunity-evidence.jsonl`
- `data/reports/phase6b-2026-08-02/career-shortlist-coverage.json`
- `data/reports/phase6b-2026-08-02/career-source-matrix.json`
- `data/reports/phase6b-2026-08-02/career-threshold-sensitivity.json`
- `data/reports/phase6b-2026-08-02/replay-manifest.json`
- `project-history/phases/phase-6/research/run_phase6b_opportunity_probe.mjs`
- `docs/research/phase6a-career-education-source-study.md`
- the current active release report;
- the current preference presets and benchmark-profile definitions; and
- the current source, lineage, country-mapping, and reproducibility conventions.

Treat the repository as authoritative where it has moved beyond these documents.

Background

Phase 6B established Opportunity Criteria as non-ranking tri-state filters with these states:

- `VERIFIED_STRONG_SIGNAL`
- `STRONG_SIGNAL_NOT_ESTABLISHED`

- INSUFFICIENT_EVIDENCE

The study approved three candidates for implementation design:

- Health and social-work employment ecosystem
- Finance and insurance employment ecosystem
- Skilled-trades or construction employment ecosystem

It held:

- Technology and software employment ecosystem
- Science and engineering employment ecosystem
- Research and academia opportunity

Technology and science/engineering already pass the minimum 60-country assessable floor, but fail the preferred benchmark-shortlist coverage gate.

They are assessable for only 15 of 20 countries in four relevant benchmark lists.

The material gaps are:

- Canada
- Japan
- South Korea
- Malta
- New Zealand

These candidates are important enough to justify a focused official-source gap-fill before implementation begins.

Phase 6B also used a P60-based strong-signal threshold. P50 and P70 sensitivity results show material country-state changes, so threshold calibration and confidence treatment must be improved before product implementation.

Objective

Strengthen the Phase 6B career-opportunity portfolio by answering four focused questions:

1. Can official national or regional sources fill the five prominent-country gaps for technology and science/engineering?
2. Can those sources be mapped reproducibly to the approved Koser occupation families without weakening construct integrity?
3. Is P60 a defensible strong-signal threshold when tested against known anchor countries and source refresh sensitivity?
4. How should observed, modelled, and supplemental evidence confidence be represented and combined?

The desired outcome is a final implementation-design portfolio that preferably includes:

- Technology and software employment ecosystem
- Science and engineering employment ecosystem
- Care-sector employment ecosystem
- Finance and insurance employment ecosystem
- Trades and construction employment ecosystem

Do not force approval. Hold any candidate that still fails the evidence gates.

Scope

This phase covers only:

- technology and software gap filling;
- science and engineering gap filling;
- threshold calibration for all Phase 6B candidates;
- confidence-tier policy;
- refinement of broad approved candidate names and internal evidence routes;
- final career-opportunity portfolio recommendation.

Research and academia should not be repaired as an employment-opportunity criterion in this phase unless an exact new labour-market source is found unexpectedly.

Instead, recommend whether it should move to the education/research ecosystem exploration as:

- Research and academic ecosystem

That future construct may describe ecosystem presence, not active academic-job availability.

Part A — Freeze the occupation-family constructs

Retain the Phase 6B constructs unless exact official data requires a truthful narrowing.

Technology and software

Preferred family:

- ISCO-08 group 25 — ICT professionals
- ISCO-08 group 35 — Information and communications technicians

Continue to exclude unrelated managers, generic professionals, electrotechnology roles, installers, and clerical ICT-adjacent roles unless explicitly justified.

The public construct should remain:

A substantial and established technology and software employment ecosystem.

It must not imply:

- current vacancies;
- salary;
- seniority fit;
- applicant hiring probability;
- visa eligibility;
- qualification recognition;
- language fit;
- or city-level demand.

Science and engineering

Preferred family:

- ISCO-08 group 21 — Science and engineering professionals
- ISCO-08 group 31 — Science and engineering associate professionals

Do not relabel this as pure engineering unless the source supports engineering-specific codes.

The public construct should remain:

A substantial and established science and engineering employment ecosystem.

Part B — Official gap-fill research

Research exact official sources for these countries:

- Canada
- Japan
- South Korea
- Malta
- New Zealand

Prioritize official statistical and labour-market authorities.

Possible source families include, but are not limited to:

Canada

- Statistics Canada employment by occupation;
- NOC-based labour-force tables;
- official downloadable CSV, API, or machine-readable tables.

Japan

- e-Stat;
- Labour Force Survey;

- official occupation tables using the Japan Standard Occupational Classification or another documented occupation taxonomy.

South Korea

- KOSIS;
- official employment by occupation tables;
- KSCO-based labour statistics.

Malta

- Eurostat Labour Force Survey;
- Malta National Statistics Office;
- official ISCO-based employment tables;
- Eurostat ICT-specialist indicators only if their construct and denominator align.

New Zealand

- Stats NZ;
- official labour-market releases;
- ANZSCO-based employment by occupation tables;
- downloadable or API-accessible official data.

Do not assume these are usable merely because they exist.

For each source freeze:

- exact table or API endpoint;
- publisher;
- dataset title;
- version or extraction date;
- reference period;
- occupation classification;
- sex dimension;
- employment unit;
- denominator availability;
- quality or break flags;
- licence and production-use terms;
- normalized-derivative rights;
- attribution;
- raw-retention policy;
- machine replay method;
- checksum or deterministic query identity.

Part C — Classification crosswalks

Create research-only, versioned crosswalks from each national classification to the approved Consider occupation families.

Examples may include:

- NOC to ISCO/Konsider technology and science-engineering families;
- JSOC to Konsider families;
- KSCO to Konsider families;
- ANZSCO to Konsider families;
- Malta/Eurostat ISCO mappings.

Each mapping must include:

- source classification and version;
- source codes;
- included codes;
- excluded adjacent codes;
- one-to-many or many-to-one treatment;
- mapping rationale;
- known ambiguity;
- expected over-inclusion or under-inclusion;
- confidence level;
- official crosswalk provenance where available.

Prefer official crosswalks.

Where no official crosswalk exists:

- do not invent an implicit mapping;
- document the gap;
- use a transparent research mapping only if the semantic correspondence is strong;
- classify the result as lower confidence;
- do not use it to assign `STRONG_SIGNAL_NOT_ESTABLISHED` unless detection completeness is defensible.

Reject a national source if the mapping becomes too broad or too arbitrary.

Part D — Metric compatibility

The current opportunity route uses:

- occupation-family employment scale; and
- occupation-family share of total employment.

For every national gap-fill source, determine whether both can be computed compatibly.

Requirements:

- numerator and denominator from the same country, period, population, and source family where possible;
- both-sex total;

- employment stock, not vacancies;
- no mixing of incompatible survey populations;
- no mixing of seasonally adjusted monthly values with annual totals without a frozen conversion policy;
- no inferred values from charts or search snippets;
- no combining stale and fresh components.

If only one dimension is available, test whether it can support:

- a positive corroborating route only; or
- no classification at all.

Do not use a one-dimensional source to establish a negative result unless the approved policy explicitly proves adequate detection.

Part E — Evidence precedence and confidence tiers

Define a research-level confidence model.

At minimum test these tiers:

HIGH_CONFIDENCE_OBSERVED_HARMONISED

Examples:

- exact observed ILOSTAT occupation data;
- harmonised Eurostat LFS data with exact ISCO mapping.

HIGH_CONFIDENCE_OBSERVED_NATIONAL

Official national observed data with a strong classification mapping and compatible scale/share calculation.

MEDIUM_CONFIDENCE_MODELLED_HARMONISED

ILO modelled estimates covering 88 countries.

MEDIUM_CONFIDENCE_OFFICIAL_CROSSWALK

Official national data requiring a documented classification crosswalk with limited ambiguity.

LOW_CONFIDENCE_SUPPLEMENTAL

Current-demand, shortage-list, employer-presence, or partial evidence that may support explanation but cannot independently establish a negative state.

The final public tri-state result remains unchanged.

Confidence is additional evidence metadata, not a fourth public state.

Define source precedence when multiple routes exist.

A reasonable starting order to test is:

1. recent harmonised observed;
2. recent official national observed with strong mapping;
3. harmonised modelled;
4. approved supplemental corroboration.

Do not automatically overwrite a valid observed result with a modelled estimate.

Record contradictions explicitly.

Part F — Multi-route classification policy

Evaluate an asymmetric rule.

VERIFIED_STRONG_SIGNAL

May be established through:

- the harmonised observed ILO route; or
- an approved official national observed route; or
- a defined combination of moderate employment evidence and credible supplemental evidence.

STRONG_SIGNAL_NOT_ESTABLISHED

May be assigned only when:

- a high- or medium-confidence employment-stock route evaluated both scale and share;
- classification coverage is sufficiently complete;
- no strong-signal rule passes;
- no major unresolved contradiction exists.

INSUFFICIENT_EVIDENCE

Must be assigned when:

- source data are missing or stale;
- classification mapping is materially ambiguous;
- only supplemental or current-demand evidence exists;
- denominator compatibility fails;
- legal reuse is unclear;
- or evidence routes materially conflict without a frozen resolution rule.

Do not lower the integrity of the negative state merely to increase coverage.

Part G — Benchmark shortlist reassessment

Regenerate the same five benchmark top-20 lists using only existing ordering criteria and existing ranking logic:

- general balanced;
- affordability-sensitive;
- safety and governance-oriented;
- career-prioritised;
- family and education-oriented.

Do not use opportunity evidence to create or reorder the lists.

For technology and science/engineering, report:

- assessable countries per list before gap fill;
- assessable countries per list after gap fill;
- which countries changed from insufficient evidence;
- their resulting state;
- evidence source and confidence;
- whether the preferred 16/20 gate passes;
- whether 18/20 or 20/20 is achieved.

The preferred target is:

- at least 16/20 in every relevant list;
- preferably 18/20 or better.

Do not force a pass by assigning weak national mappings.

Part H — Threshold calibration

The Phase 6B base rule is:

- scale $\geq$ P60 and share $\geq$ P60; or
- scale $\geq$ P80 and share $\geq$ P40; or
- share $\geq$ P80 and scale $\geq$ P40.

P50 and P70 were sensitivity scenarios.

Do not assume P60 is correct merely because it was the middle tested value.

Anchor-country validation

Create a frozen, evidence-based anchor set for each candidate.

Include:

- expected strong examples;
- expected borderline examples;
- expected not-established examples;
- countries whose classification would be surprising and require review.

Anchor selection must be justified from independent descriptive evidence or well-established ecosystem facts.

Do not tune thresholds solely to make famous countries pass.

For each candidate evaluate:

- P50, P55, P60, P65, and P70;
- number of strong signals;
- geographic distribution;
- income-group distribution;
- shortlist effects;
- anchor-country precision;
- unexpected positives;
- unexpected negatives;
- stability under removal of one source route;
- stability under latest-year versus prior-year data where available.

Raw-threshold stability

Percentiles must eventually become frozen raw thresholds for a published evidence edition.

Test whether the raw thresholds remain reasonably stable across:

- source refreshes;
- observed versus modelled distributions;
- addition of five national-source countries;
- exclusion of low-confidence routes.

If thresholds shift substantially, recommend either:

- source-family-specific thresholds;
- confidence-aware thresholds;
- a fixed reference distribution;
- or holding implementation until stability improves.

Do not create country-specific thresholds.

Part I — Review the three approved broad criteria

Reassess the approved Phase 6B candidates.

Health and social work

Determine whether the public name should be:

- Health and social-work employment ecosystem;
- Care-sector employment ecosystem; or
- another truthful broad name.

Where evidence permits, retain separate internal components for:

- human health;
- social work;
- combined section Q.

Do not call section Q healthcare alone.

Finance and insurance

Determine whether section K should remain:

- Finance and insurance employment ecosystem.

Assess whether observed business/administration occupation evidence can be shown as supplemental detail without expanding the public name beyond the source.

Do not call section K business and administration.

Skilled trades and construction

The current candidate passes through either:

- skilled-trades occupation evidence; or
- construction-industry evidence.

Decide whether this should remain one public Opportunity Criterion.

Evaluate:

1. how many countries pass through skilled trades only;
2. how many pass through construction only;
3. how many pass through both;
4. whether those groups have materially different product meaning;
5. whether the public result can clearly explain the passing route.

Choose one recommendation:

- keep one combined public criterion with visible route attribution;
- split into two future criteria;

- narrow the name;
- or hold pending clearer semantics.

Do not implement the decision in runtime code.

Part J — Research and academia disposition

Do not continue using ISIC M72 as a proxy for academic job opportunity unless a materially better exact source is discovered.

Assess whether the Phase 6B candidate should be formally moved to Phase 6C or another education/research phase under the name:

- Research and academic ecosystem

That construct may use:

- research institutions;
- university breadth;
- field-specific output;
- research intensity;
- locality clusters.

It must not imply active vacancies or applicant employability.

Record a clear disposition:

- move to education/research ecosystem study;
- hold for a new labour-market source;
- or reject as a career Opportunity Criterion.

Minimum approval gates

Technology or science/engineering may move to implementation design only if all applicable gates pass.

Coverage

- at least 60/91 assessable;
- at least 16/20 assessable in every relevant benchmark shortlist;
- preferably at least 18/20;
- geographic and income breadth remain acceptable.

Construct

- exact public name matches the occupation family;
- national crosswalks do not materially broaden the construct;
- employment stock is not presented as vacancies or hiring probability.

Negative-state integrity

- `STRONG_SIGNAL_NOT_ESTABLISHED` is assigned only with sufficiently complete evidence;
- national-source gaps do not become false negatives;
- supplemental signals cannot establish negative outcomes.

Legal and reproducibility

- exact source rights are compatible;
- data capture is deterministic;
- crosswalks are versioned;
- country mapping is explicit;
- outputs replay from retained inputs;
- source and result checksums are frozen.

Threshold

- the selected threshold is justified by anchor, sensitivity, discrimination, and stability analysis;
- no obvious systematic distortion is left unexplained.

Decision categories

Assign each candidate exactly one final disposition:

`APPROVE_FOR_IMPLEMENTATION_DESIGN`

All research gates pass.

`APPROVE_WITH_NAMING_OR_SCOPE_CHANGE`

A narrower or broader truthful construct passes.

`HOLD_SOURCE_GAP`

Coverage or source acquisition remains inadequate.

`HOLD_CROSSWALK_GAP`

Official data exist, but classification mapping is not sufficiently deterministic.

`HOLD_THRESHOLD_CALIBRATION`

Evidence exists, but the strong-signal threshold is not stable or defensible.

`HOLD_POLICY_GAP`

Multi-route precedence, confidence, or negative-state completeness remains unresolved.

MOVE_TO_EDUCATION_RESEARCH_PHASE

The construct is better represented as an ecosystem criterion than employment opportunity.

REJECT_AS_CAREER_OPPORTUNITY

The candidate cannot be made sufficiently precise or useful.

Required deliverables

Create:

1. docs/research/phase6b1-career-gap-fill-and-calibration.md
2. data/reports/phase6b1-<date>/national-source-matrix.json
3. data/reports/phase6b1-<date>/occupation-crosswalks.json
4. data/reports/phase6b1-<date>/gap-fill-country-evidence.jsonl
5. data/reports/phase6b1-<date>/career-confidence-policy.json
6. data/reports/phase6b1-<date>/multi-route-decision-policy.json
7. data/reports/phase6b1-<date>/career-shortlist-coverage-before-after.json
8. data/reports/phase6b1-<date>/career-threshold-calibration.json
9. data/reports/phase6b1-<date>/anchor-country-review.json
10. data/reports/phase6b1-<date>/broad-criterion-route-analysis.json
11. data/reports/phase6b1-<date>/final-career-opportunity-portfolio.json
12. data/reports/phase6b1-<date>/replay-manifest.json
13. deterministic research scripts and small fixtures required for replay.

The final portfolio must include:

- final candidate ID;
- final public name;
- exact construct;
- public tri-state semantics;
- evidence routes;
- confidence tiers;
- source precedence;
- assessable-country count;
- state counts;
- benchmark coverage;
- threshold policy;
- legal status;
- limitations;
- implementation-design disposition.

Explicit non-goals

Do not:

- change the active release;
- create a production release;
- change affinity scoring;
- change PCC rules;
- change the stable 91-country universe;
- add runtime Opportunity Criterion contracts;
- modify the worker production path;
- add API filters;
- add UI controls;
- add conversational behavior;
- add applicant profiles;
- add visa, licensing, salary, or qualification-recognition logic;
- implement education criteria;
- or begin implementation design beyond research recommendations.

Verification

At minimum verify:

- exact input checksums;
- deterministic replay;
- all national source captures or queries;
- crosswalk schema validity;
- exactly one gap-fill result per targeted country and candidate;
- country-code reconciliation;
- numerator and denominator compatibility;
- shortlist before/after reconciliation;
- threshold scenario regeneration;
- anchor-country assertions;
- confidence and source-precedence assertions;
- JSON/JSONL validation;
- no-diff report regeneration;
- documentation links;
- lint and formatting for research scripts;
- existing regression tests affected by documentation or history changes.

Do not claim remote CI unless it actually ran.

Stop condition

Stop after research, calibration, and the final career-opportunity portfolio recommendation.

Report:

- official national sources evaluated;
- crosswalks accepted or rejected;
- results for Canada, Japan, South Korea, Malta, and New Zealand;
- technology shortlist coverage before and after;
- science/engineering shortlist coverage before and after;
- confidence policy;
- selected threshold and rationale;
- broad-criterion naming decisions;
- research/academia disposition;
- final approved, held, moved, or rejected candidates;
- files changed;
- tests run;
- commit SHA;
- and owner decisions required before implementation design.

Do not proceed to production code, data-model changes, API, UI, releases, or conversational integration.